

UCS503 – Software Engineering

Smart City Civic Complaint and Management System

A Web-Based Platform for Civic Complaint Registration, Tracking and Resolution

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Chapter 1

Introduction

1.1 Background and Context

Urban areas generate a large number of civic complaints related to roads, street lights, sanitation, drainage, waste management, public infrastructure, and other municipal services. In a conventional complaint-management process, citizens may have limited visibility into the status of their complaints after submitting them. Complaints can also require coordination between different departments and field-level workers before the reported issue is resolved.

The Smart City Civic Complaint and Management System is proposed as a web-based platform that provides a structured process for registering, managing, tracking, and resolving civic complaints. The system connects citizens with department officers and field workers through a centralized workflow.

A citizen can register or log in to the system, submit a complaint by providing relevant details and location information, and optionally upload a photograph as supporting evidence. The system generates a unique complaint ID and maintains the complaint information for further processing.

After submission, the department officer can review and verify the complaint and assign an appropriate field worker. The field worker can view the assigned task, accept the work, update its progress, upload evidence, and mark the work as completed. The citizen can subsequently verify whether the reported issue has been resolved.

The system therefore establishes a complete complaint lifecycle from submission to resolution and citizen verification. The project also considers smart assistance features such as complaint classification, duplicate detection, and priority calculation as planned extensions to the core system.

1.1.1 Problem Statement

The existing complaint-handling process can involve several disconnected activities such as complaint registration, department assignment, field-level work, status updates, and resolution verification. Without a centralized system, it can be difficult for citizens to track complaints and for responsible personnel to maintain a consistent record of complaint progress.

The specific problem addressed by this project is the lack of a structured and traceable platform for managing the complete lifecycle of civic complaints.

The proposed system aims to provide:

- A centralized platform for registering civic complaints.
- A structured mechanism for recording complaint details and location.
- Unique identification of every submitted complaint.
- Assignment of verified complaints to appropriate department officers and field workers.
- Continuous status updates during complaint resolution.
- Uploading of photographs and work evidence.
- Citizen verification of the completed work.
- An option to reopen a complaint when the issue has not been properly resolved.
- A foundation for future smart features such as classification, duplicate detection, and priority calculation.

1.2 Project Objectives

The major objectives of the Smart City Civic Complaint and Management System are:

1. To develop a centralized web-based platform through which citizens can register and submit civic complaints.
2. To provide a structured complaint workflow covering complaint submission, verification, worker assignment, work progress, resolution, and citizen verification.
3. To maintain complaint information such as category, description, location, photographs, status, and resolution details in an organized manner.
4. To allow department officers to review complaints, verify their validity, assign field workers, and monitor complaint progress.

5. To allow field workers to view assigned tasks, update work progress, upload work evidence, and mark tasks as completed.
6. To provide citizens with complaint tracking and resolution verification functionality.
7. To design the system in an incremental manner so that advanced features such as smart classification, duplicate detection, and priority calculation can be incorporated in later stages.
8. To maintain a traceable record of the complaint lifecycle from initial submission to final closure.

Chapter 2

Methodology and Design

2.1 System Architecture

The Smart City Civic Complaint and Management System follows a modular web-based architecture in which different users interact with the system according to their responsibilities.

The major participants in the system are:

- **Citizen:** Registers or logs in, submits complaints, tracks complaint status, verifies resolution, provides feedback, and can reopen an unresolved complaint.
- **Department Officer:** Reviews and verifies complaints, assigns field workers, monitors progress, and manages complaint status.
- **Field Worker:** Views assigned tasks, accepts work, performs the required work, updates progress, uploads evidence, and marks the work as completed.
- **Administrator:** Manages users, departments, workers, and overall complaint information.

The system is designed around a central complaint-management workflow. Complaint information submitted by a citizen is processed by the system and subsequently handled by the department officer and field worker.

The general workflow is:

Citizen → Complaint Submission → Officer Verification → Worker Assignment → Work Execution → Reso

This architecture provides a clear separation of responsibilities between the different actors while maintaining a common complaint lifecycle.

2.1.1 High-Level Design

At a high level, the system can be divided into the following functional components:

1. Citizen Interface

The citizen interface provides functionality for registration/login, complaint submission, entering complaint details, providing location information, uploading photographs, tracking complaint status, and verifying resolution.

2. Complaint Management Layer

This layer manages the complaint lifecycle. It handles complaint registration, generation of complaint IDs, status management, verification, assignment, and closure.

3. Department Officer Module

The department officer module provides functionality for reviewing complaints, verifying their validity, assigning field workers, and monitoring the progress of complaints.

4. Field Worker Module

The field worker module allows workers to view assigned tasks, accept work, update progress, upload work evidence, and mark work as completed.

5. Administration Module

The administration module is responsible for managing users, departments, workers, and overall complaint monitoring.

6. Smart Assistance Layer

The proposed smart assistance layer contains planned features such as smart classification, duplicate detection, and priority calculation. These features are intended to be introduced after the core complaint workflow is stable.

Use Case Diagram

The Use Case Diagram represents the major actors of the system and the functions available to each actor. It establishes the functional boundary of the proposed system and shows the interaction between citizens, department officers, field workers, administrators, and the system.

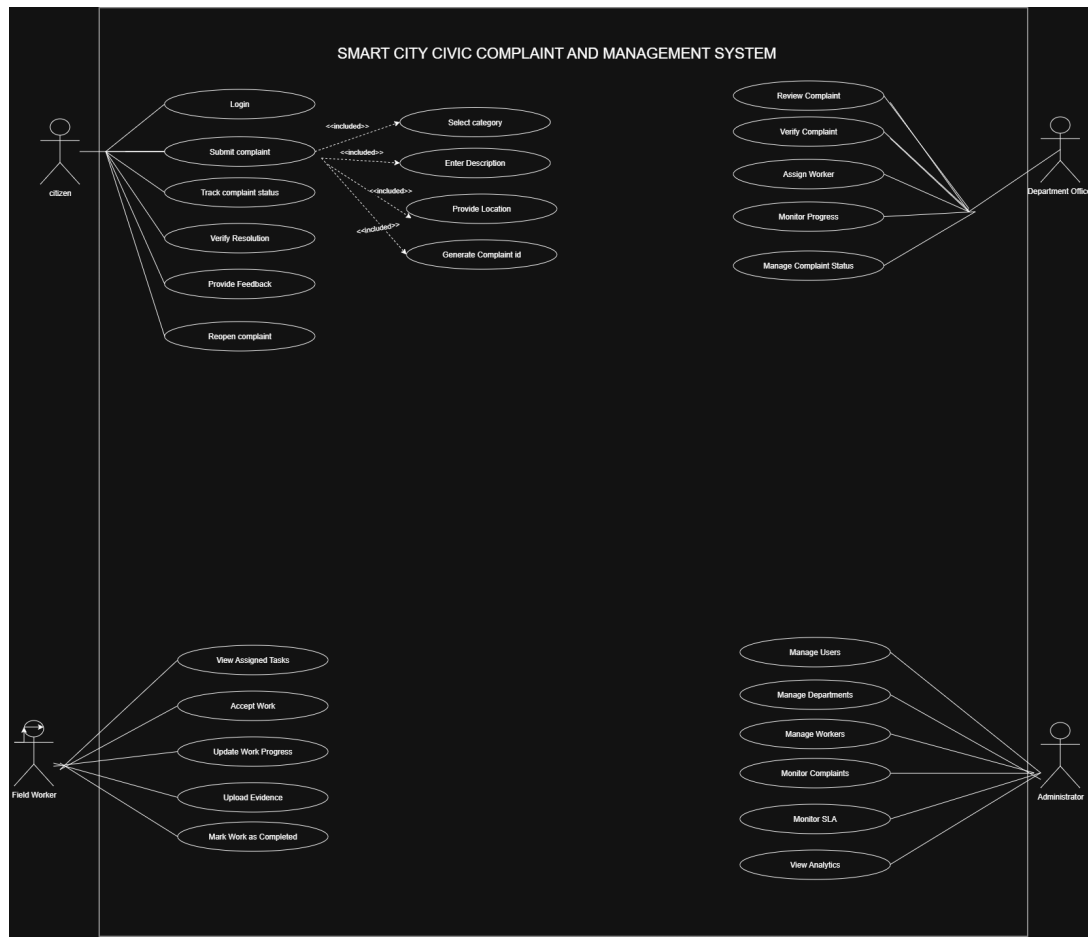


Figure 2.1: Use Case Diagram of the Smart City Civic Complaint and Management System

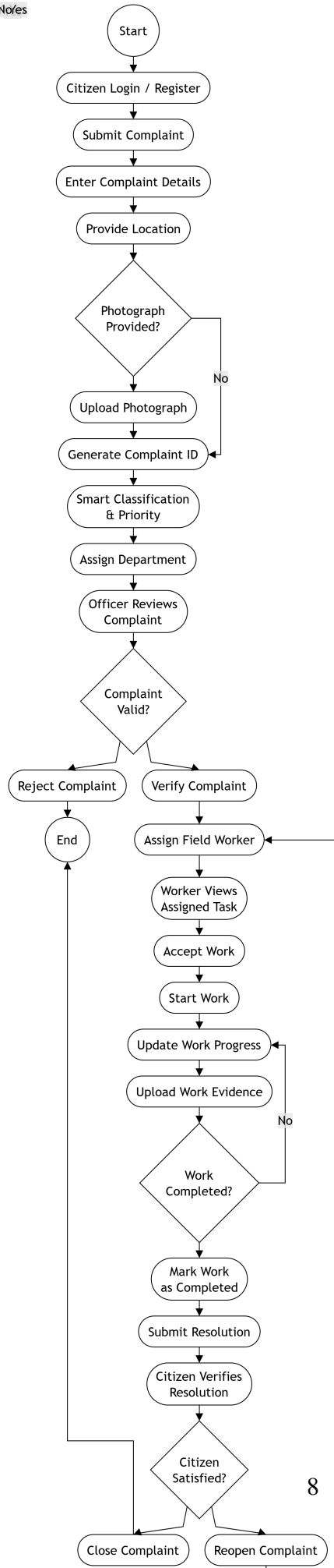
The diagram identifies four primary actors: Citizen, Department Officer, Field Worker, and Administrator. The citizen is responsible for complaint creation and verification, the department officer manages verification and assignment, the field worker performs the assigned work, and the administrator manages system-level information.

The diagram also represents the planned smart assistance features including Smart Classification, Duplicate Detection, and Priority Calculation.

Activity Diagram

The Activity Diagram represents the complete workflow of the Smart City Civic Complaint and Management System. It shows the sequence of activities starting from citizen login and complaint submission to complaint verification, worker assignment, work completion, and final citizen verification.

Notes



Sequence Diagram

The Sequence Diagram illustrates the chronological interaction between the Citizen, Smart City System, Department Officer, and Field Worker during the complaint management process. It shows how requests, responses, notifications, verification, worker assignment, progress updates, and resolution are exchanged between the different participants.

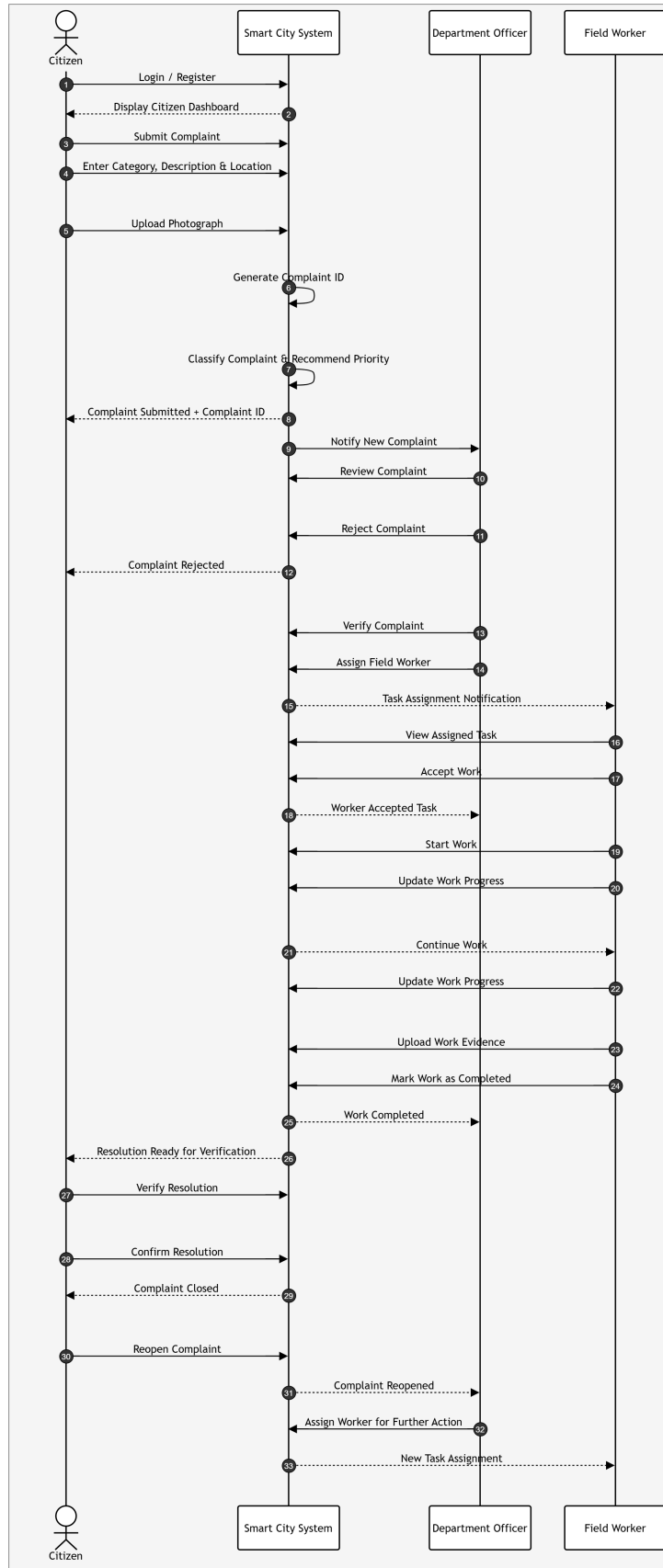


Figure 2.3: Sequence Diagram of the Smart City Civic Complaint and Management System

Swimlane Diagram

The Swimlane Diagram divides the complaint management workflow according to the responsibility of each participant. It shows which activities are performed by the Citizen, Smart City System, Department Officer, and Field Worker, making the distribution of responsibilities clear.

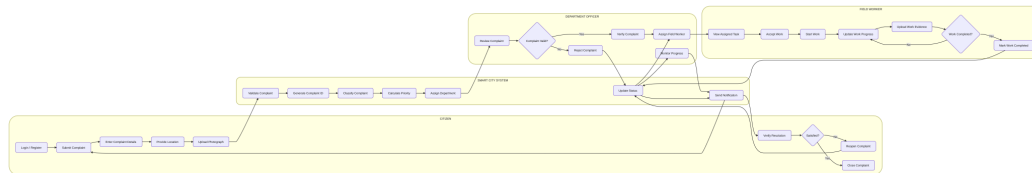


Figure 2.4: Swimlane Diagram of the Smart City Civic Complaint and Management System

Collaboration Diagram

The Collaboration Diagram represents the interaction and communication between the major objects involved in the complaint management system. It focuses on the messages exchanged between the Citizen, Smart City System, Department Officer, and Field Worker to complete the complaint lifecycle.

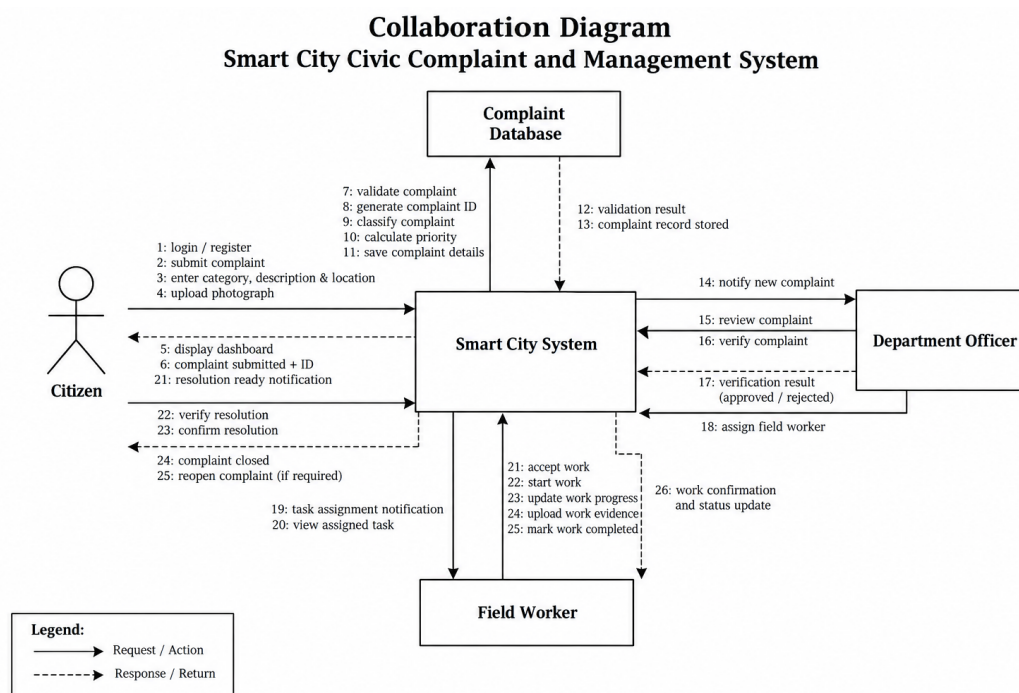


Figure 2.5: Collaboration Diagram of the Smart City Civic Complaint and Management System

2.1.2 Technical Stack

The project is being developed as a web-based application using technologies suitable for frontend development and subsequent backend integration.

Technology / Tool	Purpose
HTML	Structure and content of the web pages.
CSS	Styling, layout, spacing, typography, and visual presentation of the interface.
JavaScript	Client-side interaction and dynamic behaviour of the frontend.
VS Code	Primary development environment used for writing and managing project files.
Git	Version control for maintaining project history and tracking changes.
GitHub	Remote repository used for project collaboration and storage.
Draw.io	Used for preparing Software Engineering diagrams such as Use Case, Activity, Swimlane, and Collaboration Diagrams.

The current implementation primarily focuses on the frontend and the core user interface. Backend functionality and database integration can be incorporated in subsequent development stages.

2.2 Implementation Details

The implementation follows the functional workflow established during the system design phase.

The core implementation is organized around the complaint lifecycle. Instead of treating complaint submission, assignment, work execution, and resolution as independent processes, they are connected through complaint status and a unique complaint identifier.

The major stages of the implementation are:

1. User registration and login.
2. Complaint submission.
3. Entry of complaint details.
4. Location and photograph submission.
5. Generation of a complaint ID.
6. Complaint review and verification.
7. Assignment of a field worker.

8. Field work and progress updates.
9. Uploading of work evidence.
10. Completion of the assigned work.
11. Citizen verification of the resolution.
12. Closing or reopening of the complaint.

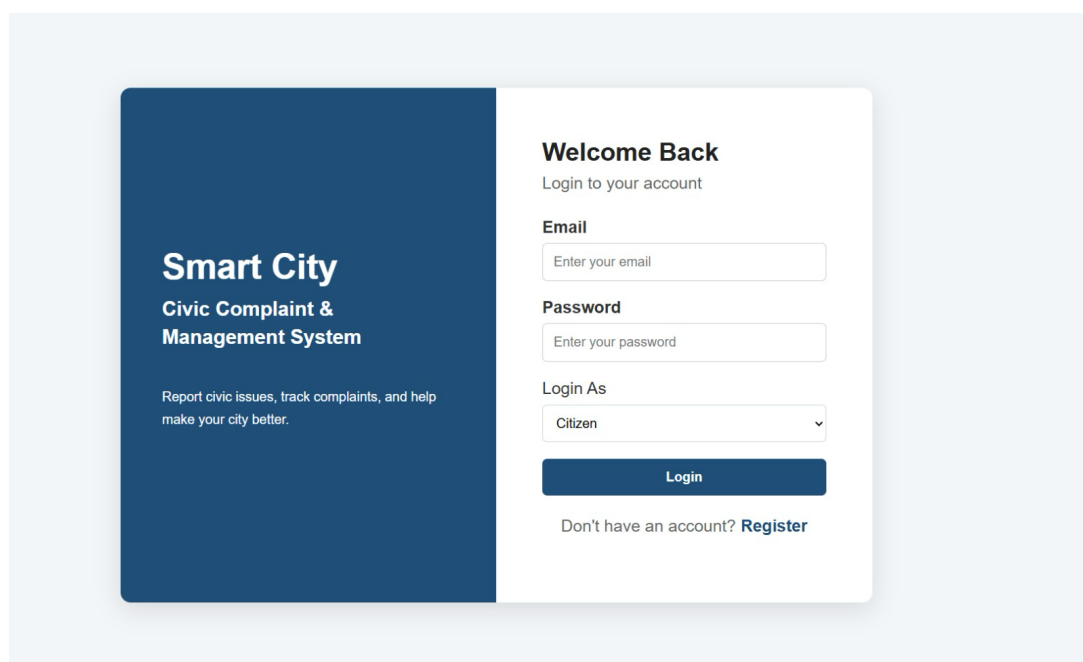
This modular approach allows individual parts of the system to be developed and tested independently while maintaining the overall complaint workflow.

2.2.1 Module 1: Core Functionality

The Core Functionality module provides the main complaint management features of the Smart City Civic Complaint and Management System. The module supports different users of the system, including Citizens, Department Officers, Field Workers, and Administrators. Each user is provided with an appropriate interface according to their role.

The module covers user authentication, role-based dashboards, complaint tracking, complaint review, field task management, and overall complaint monitoring.

User Authentication The system provides a login interface through which users can enter their email address and password. The user can also select the appropriate role, such as Citizen, Officer, or Field Worker, before logging into the system. A registration option is also provided for new users.



The image shows a user login interface for the 'Smart City Civic Complaint & Management System'. The interface is divided into two main sections: a dark blue sidebar on the left and a white login form on the right. The sidebar contains the system name and a brief description: 'Report civic issues, track complaints, and help make your city better.' The login form is titled 'Welcome Back' and 'Login to your account'. It includes input fields for 'Email' (with placeholder 'Enter your email') and 'Password' (with placeholder 'Enter your password'). Below these is a 'Login As' dropdown menu currently set to 'Citizen'. A blue 'Login' button is positioned below the dropdown. At the bottom of the form, there is a link that says 'Don't have an account? Register'.

Figure 2.6: User Login Interface

The login interface provides a simple and centralized entry point to the system. After successful authentication, the user is redirected to the dashboard corresponding to their selected role.

Citizen Dashboard The Citizen Dashboard provides an overview of the complaints submitted by the citizen. It displays the total number of complaints along with their current status, including pending, in-progress, and resolved complaints. The dashboard also provides an option to submit a new complaint and displays recent complaints for quick access.

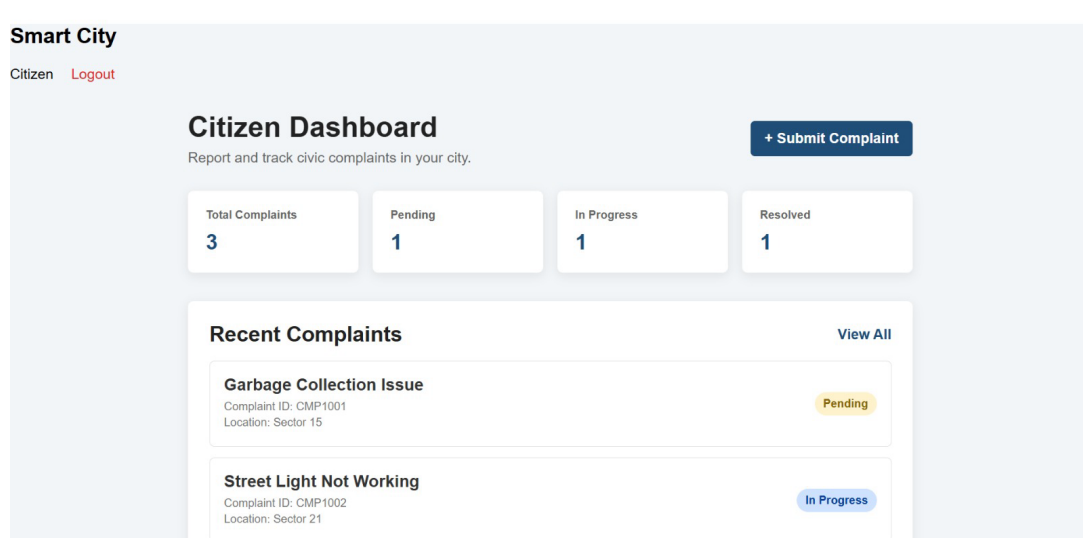


Figure 2.7: Citizen Dashboard

The dashboard allows citizens to monitor their complaints without requiring them to repeatedly contact the concerned department. The recent complaints section displays important information such as the complaint title, complaint ID, location, and current status.

Complaint Tracking Citizens can access the “My Complaints” section to view all complaints submitted through the system. Each complaint displays its complaint ID, category, location, submission date, and current status. A “View Details” option allows the citizen to inspect an individual complaint.

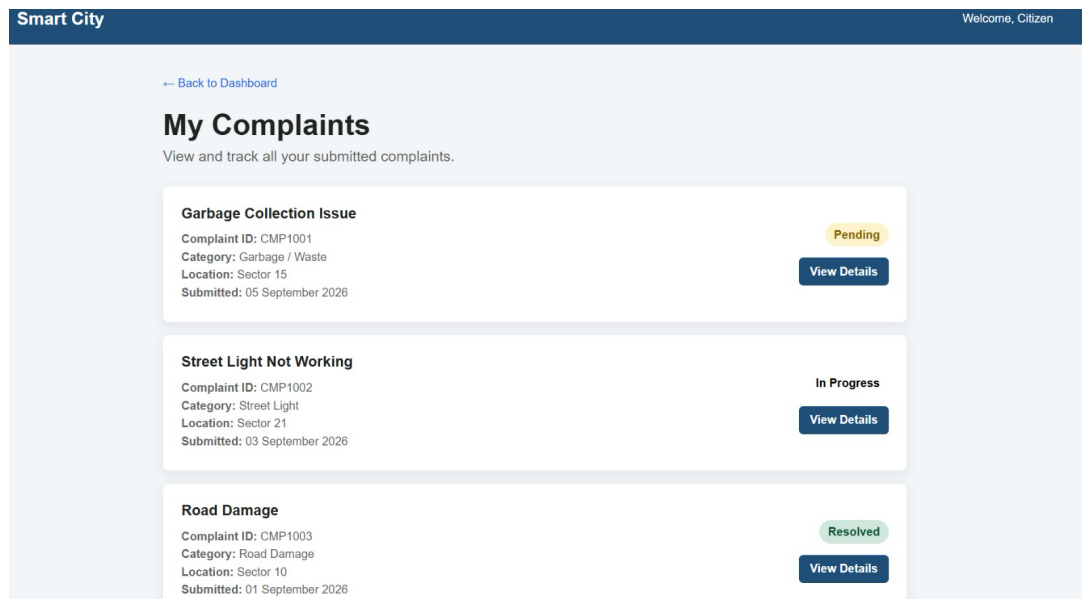


Figure 2.8: Citizen My Complaints Interface

The complaint details interface provides more detailed information about a selected complaint. It displays the complaint title, complaint ID, category, location, submission date, status, description, and complaint timeline.

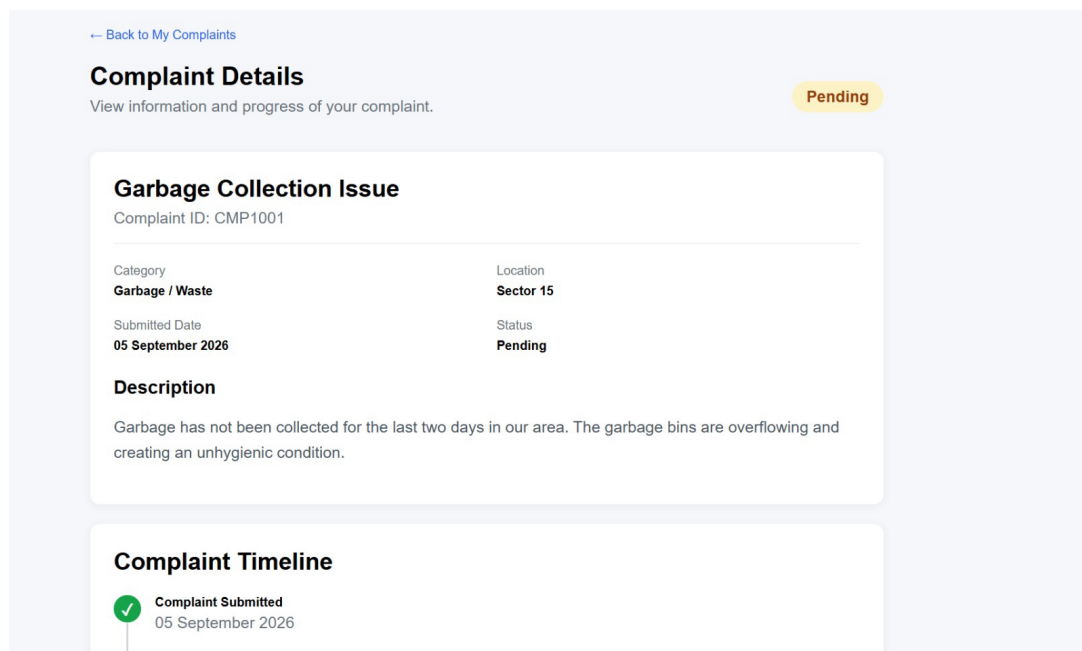


Figure 2.9: Complaint Details and Status Interface

This functionality enables citizens to track the progress of their submitted complaints and view the current state of the complaint management process.

Department Officer Dashboard The Department Officer Dashboard is designed for reviewing and managing citizen complaints. It provides a summary of total complaints, pending

verification, complaints in progress, and resolved complaints.

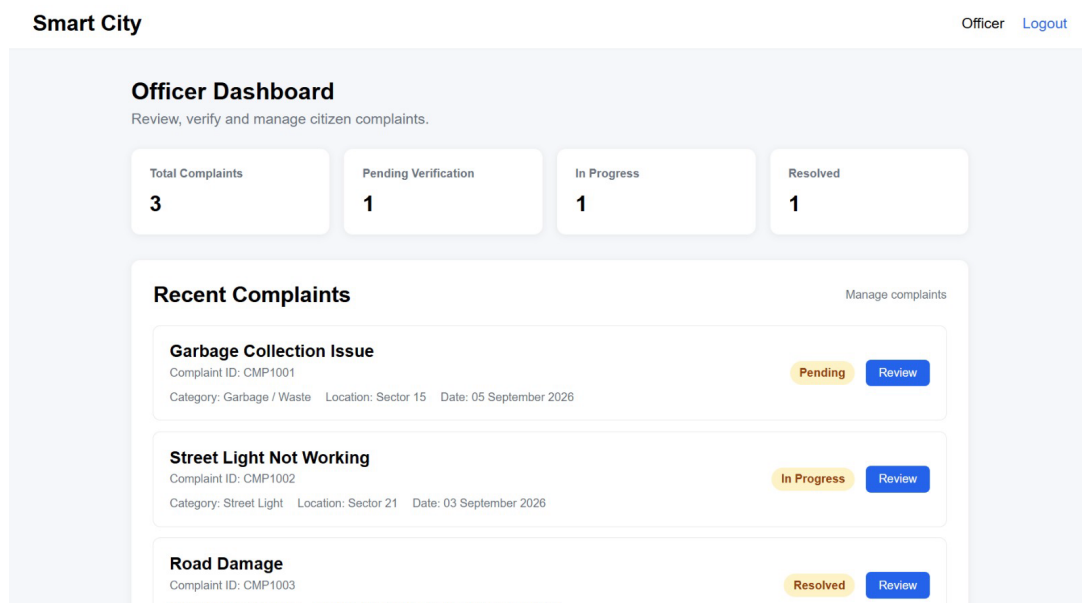


Figure 2.10: Department Officer Dashboard

The dashboard also provides a list of recent complaints. The officer can select a complaint and use the review functionality to inspect the complaint before taking further action.

Complaint Review The complaint review interface allows the Department Officer to examine the details of a submitted complaint. Information such as the complaint ID, category, location, submitted date, citizen information, email, status, and description is displayed.

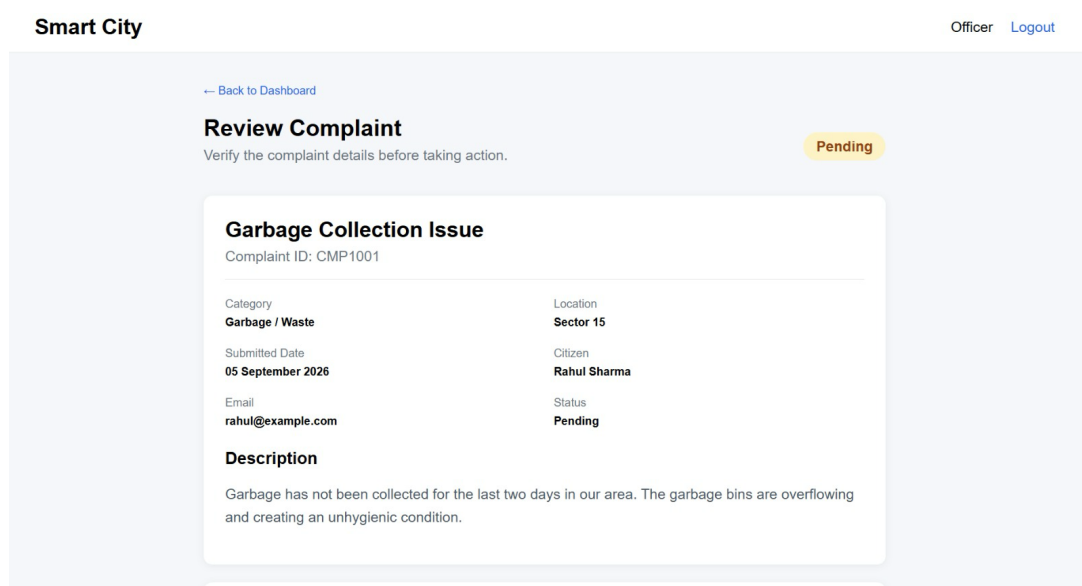


Figure 2.11: Department Officer Complaint Review Interface

This functionality provides the officer with the information required to verify and manage a

complaint before it is processed further.

Field Worker Dashboard The Field Worker Dashboard provides field workers with an overview of the complaints or tasks assigned to them. The dashboard displays the number of assigned tasks, tasks currently in progress, and completed tasks.

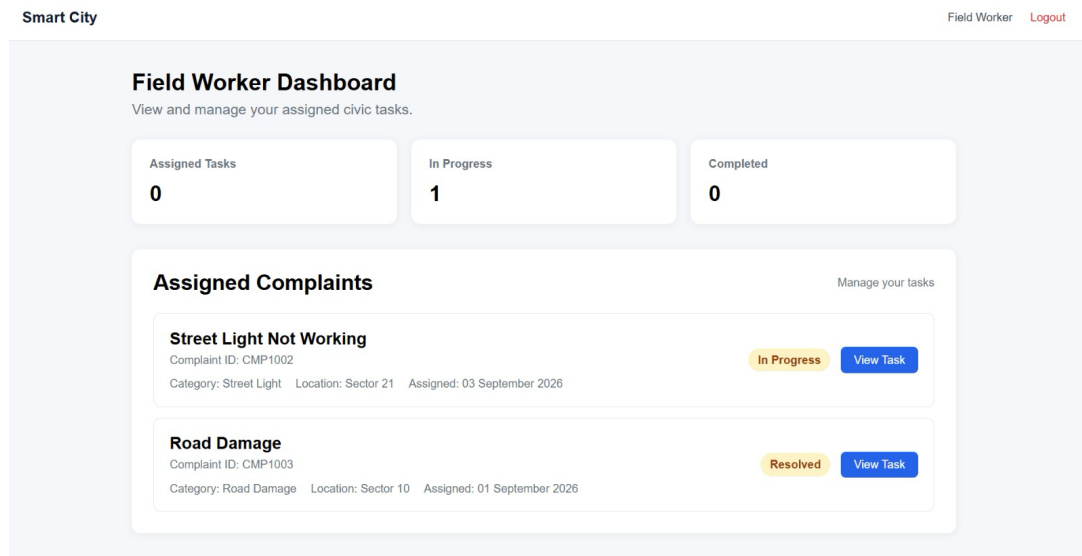


Figure 2.12: Field Worker Dashboard

The assigned complaints section provides information such as the complaint title, complaint ID, category, location, assigned date, and current status. The field worker can open an assigned task to view its complete details.

Task Management The Task Details interface provides the field worker with information about the assigned complaint. It displays the category, location, assigned date, assigned authority, current status, and complaint description.

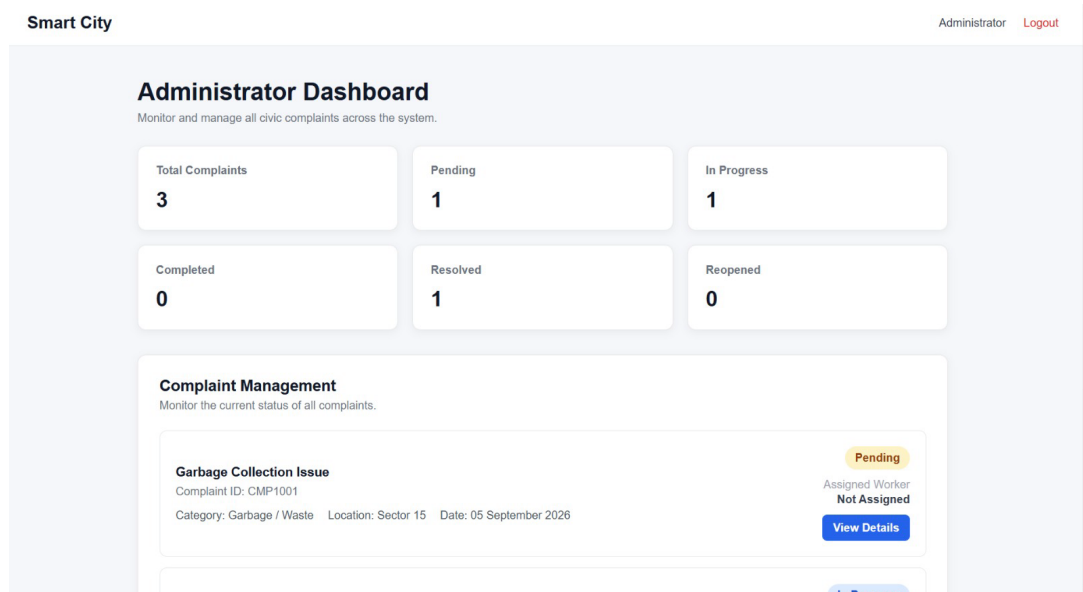


Figure 2.13: Field Worker Task Details Interface

The interface also contains a Worker Action section through which the field worker can update the progress of the assigned work. This supports the transition of a complaint from assignment to field-level resolution.

Administrator Dashboard The Administrator Dashboard provides an overall view of complaints across the system. It displays complaint statistics such as total complaints, pending complaints, complaints in progress, completed complaints, resolved complaints, and reopened complaints.

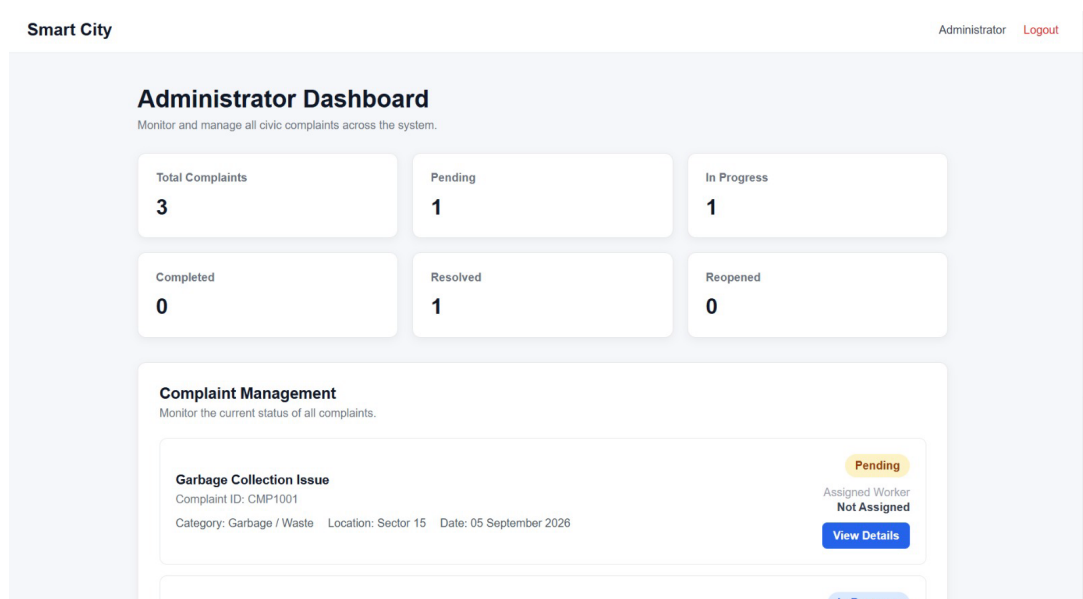


Figure 2.14: Administrator Dashboard

The complaint management section allows the administrator to monitor the current status of

complaints and view information such as complaint ID, category, location, date, and assigned worker.

Overall, the Core Functionality module establishes the main workflow of the system. A citizen can submit and track complaints, a Department Officer can review and manage complaints, a Field Worker can handle assigned tasks and update their progress, and an Administrator can monitor complaints across the system.

Core Complaint Status Flow

The core module maintains a logical sequence of complaint states:

Submitted → Under Review → Verified → Assigned → In Progress → Completed → Closed

If the complaint is rejected during verification, it follows the rejection path. Similarly, if the citizen is not satisfied with the resolution, the complaint can be reopened and sent for further action.

The status-based approach provides a traceable representation of the complaint lifecycle and allows the different system actors to understand the current state of a complaint.

Frontend Implementation

The current frontend implementation focuses on presenting the core functionality through separate interfaces for the different users.

The main frontend screens include:

- Login / Registration page.
- Citizen dashboard.
- Complaint submission form.
- Complaint tracking interface.
- Department officer complaint management interface.
- Field worker task interface.
- Work progress and evidence interface.
- Resolution verification interface.

The frontend is structured so that the interface reflects the responsibilities established in the system design. The citizen interacts mainly with complaint creation and verification, while officer and worker interfaces support the processing and resolution stages.